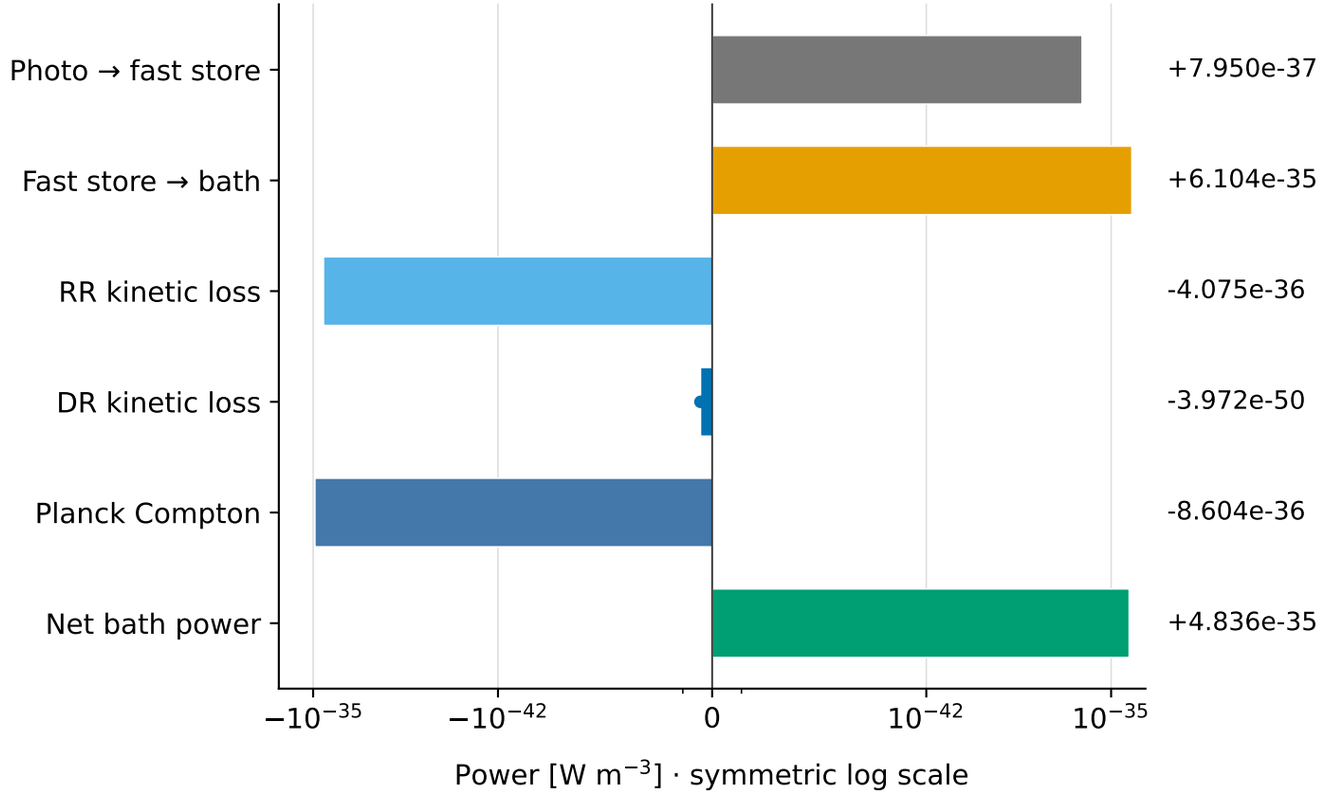
Native thermal join at the saved PW19 node

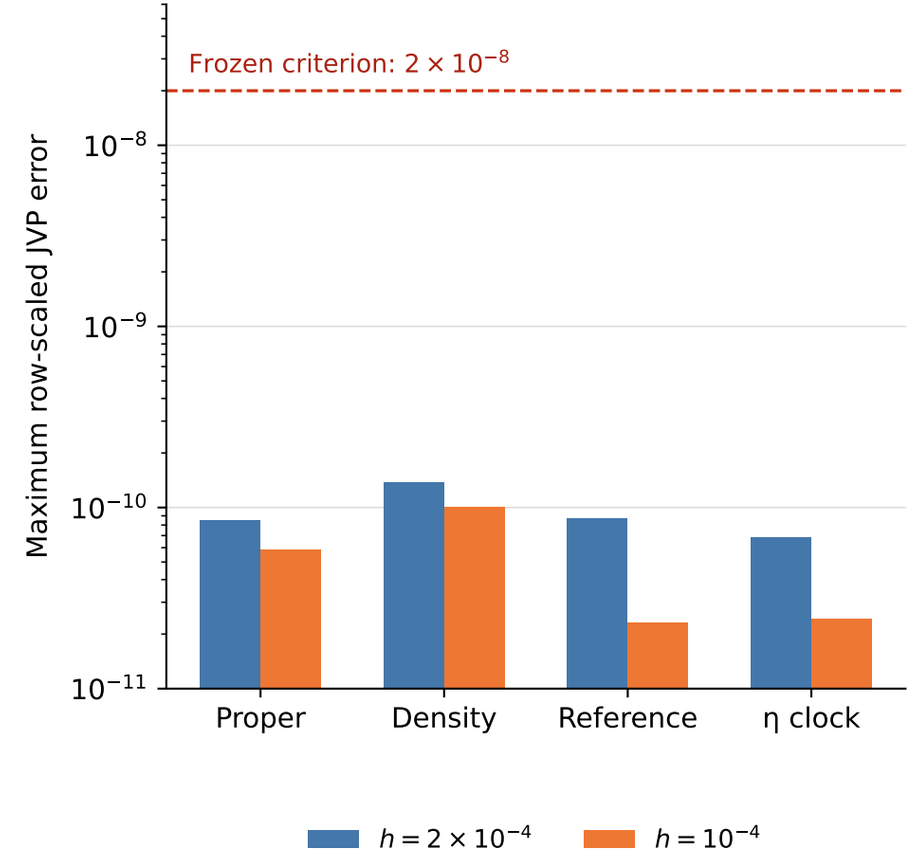
Real cached background: z = 2.479 | T = 14,919.86 K | n_b = 10.669321 m⁻³

MANUFACTURED fast pool: ζ = (q·Y)/4 = 0.21911635, w = 7.01053e-19 J/baryon

a Signed source powers



b Same-direction finite differences



Bath balance: $Q_{\mathrm{th}} = Q_A - L_{\mathrm{RR}} - L_{\mathrm{DR}} + Q_C$.

Photo primary power enters the fast store; it is separate from this bath sum.

CONDITIONAL ν = 8.160349e-18 s⁻¹: independent parameter, numerically H_snapshot.

Local source only. No trajectory or qualified physical stopping-rate claim.

Native case-A Mao2017 RR hybrid + DR cooling; isotropic Planck reference. Both FD steps use the same fixed row-unit scales.